Rockchip RKAI Developer Guide

ID: RK-KF-YF-A61

Release Version: V1.1.0

Release Date: 2026-06-06

Security Level: ☐Top-Secret ☐Secret ☐Internal ■Public

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Preface

Overview

This document presents the steps and guides for rapid verification and development of large language models and vision-language models.

Product Version

SoC	Kernel Version
RV1126B	Kernel 6.1
RK3572	Kernel 6.12

Target Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision Record

Version	Author	Date	Modification Description
V1.0.0	Cherry Chen	2025-09-30	Initial Version
V1.1.0	Cherry Chen	2026-06-06	Added RK3572 support

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1. Introduction

The Rockchip AI (RKAI) module is a rapid verification module and adaptation example for SDK integration with Large Language Models (LLM) and Vision Language Models (VLM). The source code of this module is open-source and can be found at the path: `app/rkai`.

The RKAI module supports two NPU backends:

- **RKLLM backend**: For RV1126B SoC, using RKNPU2 driver and RKLLM inference library.
- **RKNN3 backend**: For RK3572 SoC, using RKNPU3 driver and RKNN3 Session inference interface.

The two backends are controlled by the compile-time macro `RKAI_DISABLE_RKNN3`: defining this macro disables the RKNN3 backend, while leaving it undefined enables RKNN3 where supported. The actual enabled backend depends on the default configuration of the corresponding SoC SDK.

The directory structure is as follows:

```
tree  rkai -L 1
rkai
 ├──── app_conf #Configuration files and model files
 ├──── CMakeLists.txt
 ├──── common  #General code
 ├──── LICENSE
 ├──── rkai_LLM  # LLM model API source code
 ├──── rkai_VLM  # VLM model API source code
 └──── samples   # Adaptation use cases

5 directories, 2 files
```

- app_conf is the general configuration path, which includes three parts: model files, model configuration files, and model input files. This part can be directly integrated into the board for use.
- common contains general code for the RKAI module, such as `cnpy`, etc.;
- rkai_LLM is the general API for LLM deployment;
- rkai_VLM is the general API for VLM deployment;
- samples provide verification cases for LLM/Vision/VLM.

2. RV1126B Quick Verification

The SDK comes with the RKAI module pre-compiled by default and integrates the `Qwen3-0.6B` model, allowing users to quickly verify the process of large models.

> **Note**: For RK3572, see the RK3572 Quick Verification section.

2.1 LLM Quick Verification

The LLM adaptation program is `rkai_LLM_Demo`, with the following Usage:

```
Usage: rkai_LLM_Demo [options]
Options:
 --llm_model <path>   RKLLM model file
 --config    <path>   Configuration ini file
Example:
 rkai_LLM_Demo --config <path>/rkai_cfg.ini
 rkai_LLM_Demo --config <path>/rkai_cfg.ini --llm_model <your_LLM_model_path>
```

The default inference command is:

```
rkai_LLM_Demo --config <path>/rkai_cfg.ini
```

The program comes with 5 pre-set questions, from which users can select by entering numbers, or input questions directly through the terminal.

```
**********Enter the corresponding number for the following questions to get answers/ or input custom questions**********
[0] 把下面的现代文翻译成文言文: 到了春风和煦，阳光明媚的时候...
[1] 以咏梅为题目，帮我写一首古诗，要求包含梅花、白雪等元素。
[2] 上联: 江边惯看千帆过
[3] 把这句话翻译成中文: Knowledge can be acquired from many sources...
[4] 把这句话翻译成英文: RK3588是新一代高端处理器...

**************************************************************************
```

The model output consists of two parts: 1. Model response; 2. Performance.



You can also convert the model yourself and use `rkai_LLM_Demo` for model inference verification. Detailed instructions are in the Custom Models section.

2.2 VLM Quick Verification

The VLM verification workflow consists of Vision and LLM stages, which can be verified separately or together. Due to the large memory footprint of the models, it is not convenient to integrate them into the SDK. Therefore, users can quickly verify the VLM model by downloading from the following URLs.

The Vision model download link is: https://modelscope.cn/models/eliasning/Qwen3-0.6B-w4a16.rv1126b.rkllm/resolve/master/fastvlm-0.5b-fp16.rv1126b.rknn.

The LLM model download link is: https://modelscope.cn/models/eliasning/Qwen3-0.6B-w4a16.rv1126b.rkllm/resolve/master/fastvlm-0.5B-w8a8_level0.rv1126b.rkllm.

The repository name contains `Qwen3-0.6B`, while the VLM files use FastVLM 0.5B filenames. Use the filenames exactly as shown; Linux paths are case-sensitive.

Push the downloaded models to the board's `app_conf/models` directory. Then, you can use `rkai_VLM_Demo` for rapid verification. Usage is as follows:

```
Usage: rkai_VLM_Demo [options]
Options:
 --vision_model <path>   Vision RKNN model file
 --llm_model    <path>   LLM RKLLM model file
 --image_path   <path>   Input image file
 --config       <path>   Configuration ini file
Example:
 rkai_VLM_Demo --config <path>/rkai_cfg.ini
 rkai_VLM_Demo --vision_model <path>/fastvlm-0.5b-fp16.rv1126b.rknn \
   --llm_model <path>/fastvlm-0.5B-w8a8_level0.rv1126b.rkllm \
   --image_path <path>/demo.jpg \
   --config <path>/rkai_cfg.ini
```

If the downloaded models are pushed to the `app_conf/models` path, then you can use the following command for inference:

```
rkai_VLM_Demo --config <path>/rkai_cfg.ini
```

If the downloaded models are pushed to other paths, then you need to manually specify:

```
rkai_VLM_Demo --vision_model xxx.rknn \
        --llm_model xxx.rkllm \
        --image_path <path>/demo.jpg \
        --config <path>/rkai_cfg.ini
```

The model output consists of two parts: 1. Model response; 2. Performance.



3. RK3572 Quick Verification

RK3572 uses the RKNN3 backend. Model files are in `.rknn` + `.weight` paired format. Unlike RV1126B's `.rkllm` format, RK3572's LLM and VLM models both use the RKNN3 NPU interface for inference. The RK3572 SDK does not integrate the LLM/VLM models listed below by default. Before quick verification, download the model directory and push it to the board.

The RK3572-specific configuration file is `rkai_cfg_rk3572.ini`. The actual file path depends on the SDK release package. This configuration mainly differs from the RV1126B configuration in LLM/Vision model formats, image input format, `[tensor]` image placeholder format, and model/input path defaults.

3.1 LLM Model Quick Verification

RK3572's LLM model is passed as a model directory. The directory must contain the `.rknn` model file and its companion `.weight`, `.tokenizer.gguf`, and `.embed.bin` files. The program automatically detects companion files from the model directory.

RK3572 LLM model download link: https://modelscope.cn/models/cdd12138/Qwen2.5-0.5B-Instruct. Download the `Qwen2.5-0.5B-Instruct-w4a16-1-1024/` model directory from the repository and pass the directory as the `--llm_model` argument.

Verification command:

```
rkai_LLM_Demo \
  --config /usr/share/app_conf/config/rkai_cfg_rk3572.ini \
  --llm_model <path>/Qwen2.5-0.5B-Instruct-w4a16-1-1024/
```

Model directory structure example:

```
Qwen2.5-0.5B-Instruct-w4a16-1-1024/
├── llm.rknn            # LLM model file
├── llm.weight          # Weight file
├── llm.tokenizer.gguf     # Tokenizer file
└── llm.embed.bin          # Embedding file
```

Output example:

```
rkai llm version: 0.0.1
rknn3 session init success

**********Enter the corresponding number for the following questions to get answers/ or input custom questions**********
[0] 把下面的现代文翻译成文言文: 到了春风和煦，阳光明媚的时候...
[1] 以咏梅为题目，帮我写一首古诗，要求包含梅花、白雪等元素。
...

user: 0
robot: (model output)
```

```
===== RKNN3_LLM Results Summary =====

TTFT:          xxx.xxx ms
GenLen:        xx tokens
TPS:           xx.xxx tokens/s
Total:         xxx.xxx ms
LLMMemory:     xx.xxx MB

=====================================
```

3.2 VLM Model Quick Verification

RK3572's VLM model is passed as a model directory. The directory must contain both Vision and LLM `.rknn` + `.weight` file pairs, along with the LLM's companion `.tokenizer.gguf` and `.embed.bin` files. The program automatically detects companion files from the model directory.

RK3572 VLM model download link: https://modelscope.cn/models/cdd12138/Qwen3-VL-2B-Instruct-384x384. Download the `Qwen3-VL-2B-Instruct-384x384-w4a16-1-1024/` model directory from the repository and pass the directory as the `--vision_model` and `--llm_model` arguments.

Verification command:

```
rkai_VLM_Demo \
  --vision_model <path>/Qwen3-VL-2B-Instruct-384x384-w4a16-1-1024/ \
  --llm_model <path>/Qwen3-VL-2B-Instruct-384x384-w4a16-1-1024/ \
  --image_path <path>/Qwen3-VL-2B-Instruct-384x384-w4a16-1-1024/test.jpg \
  --config /usr/share/app_conf/config/rkai_cfg_rk3572.ini
```

Model directory structure example:

```
Qwen3-VL-2B-Instruct-384x384-w4a16-1-1024/
├── vision.rknn          # Vision model file
├── vision.weight        # Vision weight file
├── llm.rknn             # LLM model file
├── llm.weight           # LLM weight file
├── llm.tokenizer.gguf   # Tokenizer file
├── llm.embed.bin        # Embedding file
└── test.jpg             # Test image
```

Output example:

```
rkai vlm version: 0.0.1
vision model: 384x384 ch=3 out_elems=xxxxx
rknn3 vlm llm session init success

**********Enter the corresponding number for the following questions to get answers/ or input custom questions**********
[0] <image>What is in the image?
[1] <image>这张图片中有什么元素?
```

```
...

user: 0
robot: (model output)

===== RKNN3_VLM Results Summary =====

VisionCost:     xxx.xxx ms
TTFT:           xxx.xxx ms
GenLen:         xx tokens
TPS:            xx.xxx tokens/s
Total:          xxx.xxx ms
VisionMemory:   xx.xxx MB
LLMMemory:      xx.xxx MB
TotalMemory:    xx.xxx MB

=====================================
```

4. Custom Models

4.1 RV1126B Custom Model Configuration

RV1126B uses `rknn-llm` for model conversion. Tool download link: https://github.com/airockchip/rknn-llm. Users can convert models on their own and perform inference using `rkai_LLM_Demo` and `rkai_VLM_Demo`. If users customize their models, `rkai_cfg.ini` needs to be modified accordingly.

1. In `rkai_cfg.ini`, the `[llm_cfg]` section contains LLM parameters, including model hyperparameters that users can adjust. The parameters are as follows:

Parameter	Default Value	Description
llm_model	models/Qwen3-0.6B_W4A16_RV1126B.rkllm	Can be defined externally with `--llm_model`
max_context_len	1024	Maximum context length
max_new_tokens	128	Maximum output tokens
skip_special_token	1	Whether to skip special characters
base_domain_id	0	NPU memory domain ID, default is 0
top_k	1	Greater than or equal to 1

2. In `rkai_cfg.ini`, the `[vlm_cfg]` section contains VLM parameters, including model hyperparameters that users can adjust. The parameters are as follows:

Parameter	Default Value	Description
vision_model	models/fastvlm-0.5b-fp16.rv1126b.rknn	Can also be defined externally with `--vision_model`
llm_model	models/fastvlm-0.5B-w8a8_level0.rv1126b.rkllm	Can be defined externally with `--llm_model`
image_path	inputs/image_1024x1024.npy	Supports input jpg images, specified with `--image_path`
max_context_len	512	Maximum context length
max_new_tokens	128	Maximum output tokens
skip_special_token	1	Whether to skip special characters
base_domain_id	0	NPU memory domain ID, default is 0
top_k	1	Greater than or equal to 1

3. In `rkai_cfg.ini`, the `[prompt]` section is the system prompt, and users need to modify this part when customizing their models. The parameters are as follows:

Parameter	Default Value	Description
system_prompt	<|im_start|>system\nYou are a helpful assistant.<|im_end|>\n	Default system prompt for Qwen3
prompt_prefix	<|im_start|>user\n	Can be obtained from the model's config file
prompt_postfix	<|im_end|>\n<|im_start|>assistant\n	Can be obtained from the model's config file

4. In `rkai_cfg.ini`, the `[tensor]` section is the image placeholder for the VLM model, and users need to modify this part when customizing their models. The parameters are as follows:

Parameter	Default Value	Description
img_start	<|im_start|>	Image placeholder, obtainable from model config
img_end	<|im_end|>	Same as above
img_content	<|im_content|>	Same as above

4.2 RK3572 Custom Model Configuration

RK3572 uses `rknn3-toolkit` for model conversion. Tool download link: https://github.com/airockchip/rknn3-toolkit.

RK3572 uses `rkai_cfg_rk3572.ini`, which differs from RV1126B's `rkai_cfg.ini` in the following ways:

1. **LLM Model Files**: RK3572's LLM model requires four accompanying files: `.rknn`, `.weight`, `.tokenizer.gguf`, and `.embed.bin` (RV1126B only requires `.rkllm`).
2. **VLM Vision Model Files**: RK3572's Vision model requires both `.rknn` and `.weight` files (RV1126B only requires `.rknn`).
3. **Image Placeholders**: When using Qwen3-VL models on RK3572, the default image placeholders are:

Parameter	Default Value	Description
img_start	<|vision_start|>	Image placeholder, obtainable from model config
img_end	<|vision_end|>	Same as above
img_content	<|image_pad|>	Same as above

4. **Image Input Format**: RK3572 uses `.jpg` images for `--image_path` in the typical verification flow, and the program handles them with the built-in JPEG decoding flow.
5. **Model and Input Paths**: `vision_model`, `llm_model`, and `image_path` default to `none` in `rkai_cfg_rk3572.ini`, and are typically passed via command-line arguments `--vision_model`, `--llm_model`, and `--image_path`. `--vision_model` and `--llm_model` take model directories, not individual `.rknn` files.

RK3572 configuration example:

```
[vlm_cfg]
vision_model=none
llm_model=none
image_path=none

[tensor]
img_start=<|vision_start|>
img_end=<|vision_end|>
img_content=<|image_pad|>
```

5. Detailed API Documentation

Users can integrate applications by calling APIs themselves. The APIs are fully open-source, and the following will introduce the APIs.

5.1 LLM API

The APIs for initializing and deinitializing the LLM model are as follows:

API	**rkai_llm_init**
Description	LLM Initialization
Parameters	rkaiCtxLLM *ctx: LLM handle
	rkaiCfg *cfg: LLM parameter configuration
Return	int, 0 for success and non-zero for failure

API	**rkai_llm_release**
Description	LLM Deinitialization
Parameters	rkaiCtxLLM *ctx: LLM handle
Return	void

The APIs for LLM inference deployment are as follows:

API	**rkai_llm_run**
Description	LLM Inference
Parameters	rkaiCtxLLM *ctx: LLM handle
	rkaiCfg *cfg: LLM parameter configuration
Return	int, 0 for success and non-zero for failure

API	**rkai_llm_query_memory**
Description	Query LLM model NPU memory usage
Parameters	rkaiCtxLLM *ctx: LLM handle
Return	int64_t memory size in bytes

5.2 VLM API

5.2.1 VLM's Vision Model API

API	**rkai_vlm_vision_init**
Description	Vision Initialization, only for the RV1126B/RKNPU2 backend

API	rkai_vlm_vision_init
Parameters	const char *model_path: Vision model path, in .rknn format
	rkaiCtxVision *ctx: Vision handle
Return	int, 0 for success and non-zero for failure

API	rkai_vlm_vision_release
Description	Vision Deinitialization
Parameters	rkaiCtxVision *ctx: Vision handle
Return	int, 0 for success and non-zero for failure

APIs for Vision Inference are as follows:

API	rkai_vlm_vision_run
Description	Vision Inference
Parameters	rkaiCtxVision *ctx: Vision handle
	void *img_data: Input image buffer, in RGB888 format
	rkaiVisionToLLM *vtol: Vision-to-LLM parameters and buffer
Return	int, 0 for success and non-zero for failure

5.2.2 VLM's LLM API

API	rkai_vlm_llm_init
Description	LLM Initialization
Parameters	rkaiCtxVLM *ctx: VLM handle
	rkaiVLMCfg cfg: VLM parameter configuration
Return	int, 0 for success and non-zero for failure

API	rkai_vlm_llm_release
Description	LLM Deinitialization
Parameters	rkaiCtxVLM *ctx: VLM handle
Return	int, 0 for success and non-zero for failure

APIs for LLM Inference Deployment are as follows:

API	rkai_vlm_llm_run
Description	LLM Inference
Parameters	rkaiCtxVLM *ctx: VLM handle
	rkaiVLMCfg cfg: VLM parameter configuration
	rkaiVisionToLLM *vtol: Receives information from Vision
Return	int, 0 for success and non-zero for failure

API	rkai_vlm_llm_query_memory
Description	Query VLM LLM model NPU memory usage
Parameters	rkaiCtxVLM *ctx: VLM LLM handle
Return	int64_t memory size in bytes

5.3 RK3572 RKNN3 Additional APIs

When using the RKNN3 backend on RK3572, LLM still uses the general LLM and VLM LLM APIs above. For the Vision part, use the following RKNN3-specific APIs:

Vision RKNN3 API:

API	rkai_vlm_vision_init_rknn3
Description	Vision RKNN3 Initialization
Parameters	const char *model_path: Vision `.rknn` model path
	const char *weight_path: Vision `.weight` weight path
	rkaiCtxVisionRknn3 *ctx: Vision handle
	uint32_t core_mask: NPU core mask, 0 for auto-detection
Return	int, 0 for success and non-zero for failure

API	rkai_vlm_vision_run_rknn3
Description	Vision RKNN3 Inference (built-in resize and normalize)
Parameters	rkaiCtxVisionRknn3 *ctx: Vision handle
	unsigned char *img_data: Input image RGB888 data
	int img_width: Input image width
	int img_height: Input image height
	rkaiVisionToLLM *vtol: Vision-to-LLM transfer parameters
Return	int, 0 for success and non-zero for failure

API	rkai_vlm_vision_release_rknn3
Description	Vision RKNN3 Deinitialization
Parameters	rkaiCtxVisionRknn3 *ctx: Vision handle
Return	int, 0 for success and non-zero for failure

API	rkai_vlm_vision_query_memory_rknn3
Description	Query Vision model NPU memory usage
Parameters	rkaiCtxVisionRknn3 *ctx: Vision handle
Return	int64_t memory size in bytes